THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 10 · SET 03

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. A destarched potted plant is placed in a bell jar along with a dish containing potassium hydroxide solution and kept in sunlight for some hours. A leaf from the plant is then tested with iodine solution and does not turn blue-black. This shows that:

A. light is necessary for photosynthesis
B. chlorophyll is necessary for photosynthesis
C. carbon dioxide is necessary for photosynthesis
D. oxygen is necessary for photosynthesis

Ans: C KOH absorbs CO_2 in the jar; no starch forms in the leaf, so CO_2 is required.

2. A patient has had a large part of his small intestine surgically removed. Which of the following functions of his digestive system will be affected most?

A. Absorption of digested food
B. Churning of food
C. Digestion of proteins by pepsin
D. Secretion of saliva

Ans: A Villi of the small intestine absorb digested food; removal of the intestine reduces absorption.

3. Ravi, a 15-year-old boy, is much shorter than other boys of his age though he is otherwise healthy. A doctor suspects a deficiency of a hormone secreted by:

A. thyroid gland
B. adrenal gland
C. pituitary gland
D. pancreas

Ans: C Growth hormone is secreted by the pituitary gland.

4. The correct sequence in which an impulse travels through a single neuron is:

A. axon → cell body → dendrite → nerve ending
B. dendrite → cell body → axon → nerve ending
C. cell body → dendrite → nerve ending → axon
D. nerve ending → axon → cell body → dendrite

Ans: B Dendrites receive the signal, which passes through the cell body and the axon to the nerve ending.

5. In human males the testes lie outside the abdominal cavity in the scrotum because:

A. sperm formation requires a temperature lower than the normal body temperature
B. they need more space to grow
C. they are secretion glands which need air
D. they are protected by the scrotum from infection

Ans: A Sperm production needs a temperature about 2–3 °C lower than the body temperature.

6. Which of the following traits can be inherited by a child from its parents?

A. Sun-tanned skin
B. Strong muscles developed by exercise
C. Eye colour
D. A scar on the face

Ans: C Only traits controlled by genes, such as eye colour, are inherited; the others are acquired.

7. In the food chain grass → insect → frog → snake → eagle, the secondary consumer is:

A. insect
B. frog
C. snake
D. eagle

Ans: B Grass is the producer, insect the primary, frog the secondary, snake the tertiary and eagle the quaternary consumer.

8. **Assertion (A):** Reflex actions are very quick responses to stimuli.
Reason (R): In a reflex action the impulse is processed in the spinal cord and does not have to travel to the brain and back.

Ans: A The shorter pathway saves time.

9. **Assertion (A):** The flow of energy in a food chain is unidirectional.
Reason (R): The energy captured by autotrophs does not revert to the sun and is lost as heat at every trophic level.

Ans: A Energy cannot be recycled, so the flow is one-way.

10. Name two waste products that plants eliminate. By what methods do they get rid of them?

Oxygen/carbon dioxide, excess water, gums, resins, latex (any two, 1). Diffusion through stomata, storing in leaves/bark that fall off, secreting into the soil, storing in old xylem (any two methods, 1).

11. Name the part of the female reproductive system in which (i) fertilisation takes place and (ii) the embryo is implanted.

(i) Fallopian tube/oviduct (1); (ii) uterus (1).

OR

Write two functions of the testes in a human male.

Produce sperms (1) and secrete the male hormone testosterone (1).

12. A pure tall pea plant is crossed with a pure dwarf pea plant and all the F_1 plants are tall. Which trait is dominant? Write the genotype of the F_1 plants.

Tallness is dominant (1). F_1 genotype: Tt (heterozygous) (1).

13. Explain the mechanism of inhalation in human beings with reference to the diaphragm and the rib cage. How does the air pressure change inside the chest during inhalation?

The intercostal muscles contract and lift the ribs up and out; the diaphragm contracts and flattens (1½). The chest cavity expands, the pressure inside falls below atmospheric pressure and air rushes into the lungs (1½).

14. What are trophic levels? Why is the number of trophic levels in a food chain generally limited to four or five?

Trophic levels are the steps/feeding levels of a food chain (1). Only about 10% of the energy passes to the next level, so the energy available becomes too little after 3–4 transfers to support another level (2).

15. In Rohan's family, free earlobes (F) are dominant over attached earlobes (f). Rohan's father has free earlobes, his mother has attached earlobes, and Rohan himself has attached earlobes.

(a) What is the genotype of Rohan's mother?

ff (1).

(b) What is the genotype of Rohan's father? Give a reason.

Ff; Rohan got one f allele from each parent, so the father must carry f and, having free lobes, he is Ff (1).

(c) Draw a Punnett square for the cross between Rohan's parents and find the chance that their next child has attached earlobes.

Ff × ff gives Ff, Ff, ff, ff (1); probability of attached earlobes = 2/4 = 50% (1).

OR

(d) Explain with an example how a trait may be inherited but may not be expressed in an individual.

> A recessive trait is expressed only when two copies of the allele are present; in a heterozygous (Ff) person the dominant trait is expressed and the recessive one is only carried (2).

16. (i) Draw a labelled diagram of a neuron. (ii) Describe how an impulse is transmitted from one neuron to another at a synapse. (iii) State one function each of the cerebrum, cerebellum and medulla.

> (i) Labelled neuron: dendrites, cell body (with nucleus), axon, nerve endings (1½). (ii) Electrical impulse reaches the end of the axon → release of a chemical (neurotransmitter) → crosses the synapse → starts a new electrical impulse in the dendrite of the next neuron (1½). (iii) Cerebrum: thinking, memory, voluntary action; cerebellum: balance and posture; medulla: involuntary actions such as breathing, heartbeat (2).

OR

(i) Describe binary fission, budding and fragmentation with one example of each. (ii) What is regeneration? Name an organism that shows it.

> (i) Binary fission – the parent divides into two equal daughters, e.g. Amoeba; budding – a bud grows on the parent and detaches, e.g. Hydra/yeast; fragmentation – the body breaks into pieces each growing into a new individual, e.g. Spirogyra (3). (ii) Regeneration is the ability of an organism to grow back lost parts (or a whole organism from a piece) by specialised cells; e.g. Planaria (2).

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. In the reaction $2CuO + C \rightarrow 2Cu + CO_2$, the substance that is oxidised is:

A. CuO B. C C. Cu D. CO_2

> **Ans: B** Carbon gains oxygen to form CO_2, so it is oxidised; CuO loses oxygen and is reduced.

18. White silver chloride kept in a china dish turns grey when left in sunlight. This change represents:

A. a photolytic decomposition reaction B. a thermal decomposition reaction
C. a displacement reaction D. a neutralisation reaction

> **Ans: A** $2AgCl \rightarrow 2Ag + Cl_2$ in the presence of sunlight.

19. The number of molecules of water of crystallisation in washing soda is:

A. 2 B. 5 C. 7 D. 10

> **Ans: D** Washing soda is $Na_2CO_3 \cdot 10H_2O$.

20. A zinc ore of the type $ZnCO_3$ is first converted to zinc oxide by a process called:

A. roasting B. calcination
C. smelting D. froth flotation

> **Ans: B** Carbonate ores are heated strongly in limited or no air (calcination): $ZnCO_3 \rightarrow ZnO + CO_2$.

21. The alcohols CH_3OH, C_2H_5OH and C_3H_7OH form a homologous series. The next member of the series is:

A. C_3H_8O B. C_4H_9OH C. C_4H_{10} D. $C_5H_{11}OH$

> **Ans: B** Each member differs from the previous by $–CH_2–$.

22. **Assertion (A):** Ionic compounds have high melting and boiling points.
Reason (R): There are strong electrostatic forces of attraction between the oppositely charged ions, and a large amount of energy is required to break them.

> **Ans: A** R is the correct reason for A.

23. When green crystals of ferrous sulphate are heated, the colour changes and gases with a characteristic smell are evolved. State the colour change and write the chemical equation.

> Green → brown (1). $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (1); the gases smell of burning sulphur.

24. A person suffering from acidity is given baking soda. Write the chemical equation for the action of baking soda on stomach acid and state why it is a safe antacid.

> $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$ (1). It is a mild base and does not make the stomach too alkaline (1).

25. (a) Why do acids conduct electricity in aqueous solution but not in dry form? (b) How would you test the gas evolved when an acid reacts with a metal carbonate? Write the equation involved for the test. (c) What is the role of water in showing acidic character?

> (a) Acids give H^+ ions only in water (1). (b) Pass the gas through lime water – it turns milky: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ (1). (c) Water allows the ionisation of the acid to give H^+/H_3O^+ (1).

OR

How is washing soda obtained from sodium chloride? Write the chemical equations involved.

> $NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$ (1); $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ on heating (1); $Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$ (recrystallisation) (1).

26. Write the name and the structural formula of: (a) the alcohol having three carbon atoms, (b) the carboxylic acid having two carbon atoms, (c) the ketone having three carbon atoms.

> (a) Propan-1-ol, $CH_3–CH_2–CH_2–OH$ (1). (b) Ethanoic acid, $CH_3–COOH$ (1). (c) Propanone, $CH_3–CO–CH_3$ (1).

27. Manish noticed that a silver spoon kept in the kitchen has turned black and a copper vessel kept in a humid place has become coated with a green layer, whereas gold ornaments remain shiny. His teacher explained that these changes are examples of corrosion.

(a) Name the black coating on silver and the green coating on copper.

> Silver sulphide (Ag_2S) – black; basic copper carbonate – green (½ + ½).

(b) Why does gold not corrode?

> Gold is a very unreactive metal, low in the reactivity series (1).

(c) Name the two conditions necessary for the rusting of iron. Give one method to prevent it.

> Presence of air (oxygen) and moisture (1); galvanisation/painting/oiling (1).

OR

(d) Why is aluminium not corroded easily although it is more reactive than iron?

> A thin, tough layer of aluminium oxide forms on its surface and protects the metal from further reaction (2).

28. (a) Explain the cleansing action of soaps with the help of a diagram of a micelle. (b) Why do soaps not form lather with hard water? (c) State two advantages of detergents over soaps.

> (a) Soap molecule: ionic head (hydrophilic) and long hydrocarbon tail (hydrophobic); in water the tails surround the oily dirt and the heads face water forming a micelle which is washed away (2); labelled diagram (1). (b) Soap reacts with Ca^{2+}/Mg^{2+} salts to form insoluble scum (1). (c) Detergents work in hard water; their calcium/magnesium salts are soluble (any two, 1).

OR

(a) What is rancidity? State two ways of preventing it. (b) What is corrosion? Write the equation for the rusting of iron in words. (c) Balance: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$.

> (a) Oxidation of fats/oils in food leaving a bad smell and taste (1); storing in airtight containers, adding antioxidants, flushing with nitrogen (any two, 1). (b) Slow eating away of metals by air, moisture or

chemicals (1); iron + oxygen + water → hydrated iron(III) oxide (rust) (1). (c) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ (1).

SECTION C — PHYSICS — 11 questions · 25 marks

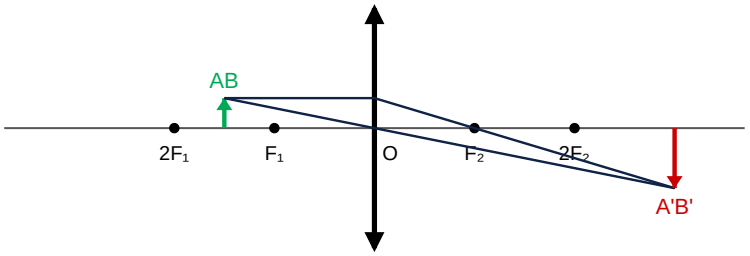
Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

29. A truck driver uses a rear-view mirror that always gives an erect, diminished image of the traffic behind. The mirror is:

A. convex
B. concave
C. plane
D. a concave mirror used with the object beyond C

Ans: A A convex mirror always forms a virtual, erect, diminished image and has a wide field of view.

30. A convex lens of focal length 10 cm forms an image of an object placed 15 cm in front of it, as shown. The image is formed at a distance of:



A. 15 cm, real and same size
B. 30 cm, real, inverted and magnified
C. 30 cm, virtual and erect
D. 7.5 cm, real and diminished

Ans: B $1/v = 1/10 - 1/15 = 1/30 \rightarrow v = 30$ cm; $m = -2$: real, inverted and magnified.

31. Danger signals are red in colour because red light:

A. is scattered the most by smoke and fog
B. has the shortest wavelength
C. is scattered the least by smoke and fog and so can be seen from far
D. has the maximum speed in air

Ans: C Red has the longest wavelength among visible light and is scattered the least.

32. The SI unit of resistivity of a material is:

A. ohm B. ohm per metre C. ohm metre D. siemens

Ans: C $\rho = RA/l$ has unit $\Omega \cdot m^2/m = \Omega$ m.

33. **Assertion (A):** Magnetic field lines never intersect each other.
Reason (R): If two field lines intersected, there would be two directions of the magnetic field at the point of intersection, which is not possible.

Ans: A R explains A.

34. What is meant by the near point of a normal human eye? Give its value.

The minimum distance at which the eye can see objects clearly without strain (1); about 25 cm (1).

OR

Why do we see the Sun about two minutes before the actual sunrise?

Atmospheric refraction: light from the Sun bends towards the normal as it passes through layers of increasing density, so the Sun appears above the horizon before it actually rises (2).

35. An object 4 cm high is placed 20 cm in front of a convex mirror of focal length 20 cm. Find the position, nature and size of the image.

u = −20, f = +20: 1/v = 1/f − 1/u = 1/20 + 1/20 = 1/10 → v = +10 cm (1). m = −v/u = +0.5 (1); h′ = 2 cm; image virtual, erect, diminished and behind the mirror (1).

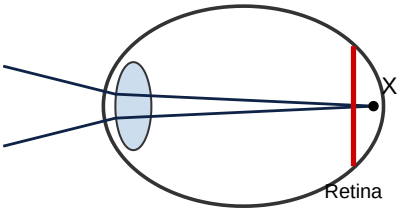
36. An electric heater of resistance 44 Ω is connected to a 220 V supply. Calculate (a) the current drawn, (b) the power of the heater and (c) the heat produced in 10 minutes.

(a) I = V/R = 5 A (1). (b) P = VI = 1100 W (1). (c) H = Pt = 1100 × 600 = 6.6×10^5 J (1).

37. Describe an activity to show that a current-carrying conductor placed in a magnetic field experiences a force. How can the direction of the force be reversed?

A small aluminium rod is suspended horizontally between the poles of a horseshoe magnet and connected to a battery through a key (1). On passing current the rod is displaced (1). Reversing the current or the direction of the magnetic field reverses the direction of the force (1).

38. Ria's grandfather finds that he can see the newspaper clearly only when it is held at arm's length (about 50 cm). He cannot read it comfortably at the normal reading distance of 25 cm, though his distant vision is perfect. His doctor prescribes spectacles.



Parallel rays would meet at X, behind the retina

(a) Name the defect of vision from which he suffers.

Hypermetropia (long-sightedness) (1).

(b) State one cause of this defect.

The focal length of the eye lens is too long or the eyeball is too short, so the image of a nearby object forms behind the retina (1).

(c) Find the power of the lens he should use to read at 25 cm.

u = −25 cm, v = −50 cm; 1/f = 1/v − 1/u = −1/50 + 1/25 = 1/50 → f = +50 cm; P = +2 D (2).

OR

(d) What is a bifocal lens? Why is it prescribed for some elderly people?

A lens with the upper part concave (distant vision) and lower part convex (near vision) (1); used when a person has both myopia and hypermetropia/presbyopia (1).

39. (a) Derive the formula for the equivalent resistance of two resistors R_1 and R_2 connected in parallel. (b) Two resistors of 6 Ω and 12 Ω are connected in parallel across a 12 V battery. Find the current through each resistor and the total current.

(a) V is the same across both, $I = I_1 + I_2$ (1); $I_1 = V/R_1$, $I_2 = V/R_2$, so $V/R = V/R_1 + V/R_2 \rightarrow 1/R = 1/R_1 + 1/R_2$ (2). (b) $I_1 = 12/6 = 2$ A (1), $I_2 = 12/12 = 1$ A (½), I = 3 A (½).

OR

A 3 cm tall object is placed 12 cm in front of a concave mirror of focal length 8 cm. (a) Find the position and size of the image. (b) Draw the ray diagram. (c) State the nature of the image.

u = −12, f = −8: 1/v = −1/8 + 1/12 = −1/24 → v = −24 cm (1½). m = −v/u = −2 (1); h′ = −6 cm (½). (b) Correct diagram with two principal rays (1). (c) Real, inverted and magnified, 24 cm in front of the mirror (1).